AMD DEVELOPER HACKATHON: ACT II — TRACK 2

STYLEFORGE

One clip. Four voices. A Gemma that learned tone on AMD silicon.

RL fine-tuned Gemma 3 4B

Trained on AMD Instinct MI300X

Ships in a CPU-only container

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The Challenge: Four Voices, One Judge

`formal`

News-wire precision. Zero jokes, zero slang.

`sarcastic`

Dry, deadpan irony. Never labeled, always felt.

`humorous_tech`

Funny via bugs, deploys, APIs and rollbacks.

`humorous_non_tech`

Funny for everyone. No tech allowed.

Every caption is scored by an LLM judge on two axes:

accuracy (0-1)

faithful to the clip

style match (0-1)

nails the requested tone

hidden eval set

generalize, don't overfit

10-minute wall

containerized, CPU, no mercy

Everyone Prompts. We Train.

THE FIELD (35+ visible teams)

- Prompt a stock VLM and hope
- Tone control via instructions only
- sarcastic vs humorous blurs together — exactly what the judge penalizes

STYLEFORGE

- DPO fine-tune Gemma on judge-ranked caption pairs (RLAIF)
- Train against the same signal the leaderboard scores
- Chosen by studying ACT I's winners: the grand prize went to an RL project; fine-tuning winners paired technique with a story

Tone is trained, not prompted

DPO on judge-ranked caption pairs — optimized against the exact signal the leaderboard scores.

Architecture: Eyes, Voice, Taste

INPUT
clip.mp4
30s-2min, UHD

→

INGEST
ffmpeg 1-pass frames
+ Whisper transcript

→

EYES
Kimi K2.6 vision dense description
(Fireworks API)

→

VOICE
Gemma 3 4B — TUNED
K captions x 4 styles
CPU, in-container

→

TASTE
LLM judge reranks best caption per style

Two-stage by design: frontier eyes, trained voice. Vision never enters the training loop — the tuned model does one narrow thing perfectly: tone.

Survives judge-time: no live GPU endpoint	Every style guaranteed: 3-tier fallback ladder	No token penalty in Track 2 - BoN is free quality

The RLAIF Data Factory

300	1,200	7,200	7,200
synthetic scenes, maximally diverse	scene x style training cells	candidate captions by Kimi K2.6	judge scores: accuracy + tone

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scene description → 6 candidates per style → judge scores each → best vs worst become a DPO preference pair (quality-gap filtered)
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DPO teaches Gemma: “write like the winner, not like the loser” — a thousand times over, across every style and scene type. Judge scores are cached, resumable, and cost-logged.

Fine-Tuned on AMD Instinct MI300X

THE STACK

- 1x MI300X (192 GB HBM3) — AMD Developer Cloud
- ROCm 7.2.4 + PyTorch 2.10 (rocm/pytorch image)
- TRL: rejection-sampling SFT warm start, merged, then DPO
- LoRA r=32, bf16 — no quantization needed to train
- Merge → GGUF Q4_K_M (~2.7 GB) for CPU serving

$1.99/hr

single-GPU droplet — credits, not cash

2.8 samples/sec

DPO throughput on a 4B LoRA

~$4

total training cost, rehearsal included

The tuned Gemma ships inside the submission container and runs on CPU — zero GPUs, zero live endpoints, zero excuses at judge time.

Built to Survive the 10-Minute Wall

SIGKILL-PROOF OUTPUT

A valid results.json is written at startup and atomically upgraded per task. The wall can kill us mid-clip — we still score on everything finished.

DEGRADE, NEVER DIE

Best-of-N → single-shot → canned fallback, per task, per style. A dead URL, a 500ing model, malformed tasks — nothing sinks the run.

HOSTILE-INPUT HARDENED

Download deadlines beat trickling servers. Single-pass UHD frame extraction. Whisper loads once. Every clip budgeted against the clock.

TESTED LIKE WE MEAN IT

11-test suite incl. an organizer-contract simulation; multi-agent adversarial code review — 8 verified findings found and fixed.

The Eval: Measured Honestly

30 unseen scenes x 4 styles x 4 arms, scored by gpt-oss-120b (no model grades its own family)

arm	formal	sarcastic	hum_tech	hum_non_tech	ALL
Kimi K2.6 prompted	8.92	7.72	8.23	7.87	**8.18**
Gemma 4B base	7.88	6.78	8.18	8.02	**7.72**
Gemma tuned v1	7.90	7.32	7.82	7.97	**7.75**
Gemma tuned v2	8.40	7.13	7.93	7.83	**7.83**
Gemma tuned v3	8.00	7.28	8.22	8.12	**7.90**
v3 vs base	+0.12	+0.50	+0.04	+0.10	**+0.18**

4B matches the frontier on accuracy

After two targeted DPO rounds our 4B scores 7.62 on factual accuracy - identical to the ~1T-class Kimi (7.62). Round 1: sarcastic +0.54. Round 2: formal +0.50.

Round 3: green across the board

Style-conditional pair weights (accuracy-heavy for formal, tone-heavy for humor) — v3 beats base on ALL FOUR styles. Three rounds, three diagnosed targets, three hits.

And it beats the frontier at humor

v3 outscores the ~1T Kimi on humorous_non_tech (8.12 vs 7.87) and ties humorous_tech. The frontier keeps the overall slot - so bon ships as default, v3 rides in-container.

We ship the winner — and the trained Gemma rides along: measured, not mythologized.

Gemma, Three Ways

THE VOICE — SHIPPED

DPO fine-tuned Gemma 3 4B writes every caption. Quantized to GGUF, it lives inside the container on CPU. We didn't prompt Gemma. We taught it.

THE EYES — DEMONSTRATED

Gemma 3 27B (vision) served by vLLM on the same MI300X: the full pipeline runs all-Gemma on AMD silicon, end to end, on camera.

THE PROOF — MEASURED

Training curves, before/after held-out evals, and an independent judge keeping us honest about our own model.

Deepest possible "use of Gemma" in this track: Gemma is the trained artifact — fine-tuned on AMD, shipped in the box.

STYLEFORGE

The only Track 2 entry where the model learned its voice.

Code	`github.com/SankarSubbayya/styleforge`
Demo	`sankarsubbayya.github.io/styleforge`
Image	`ghcr.io/sankarsubbayya/styleforge:latest`

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